Lifetime Sleep Duration Predicts Cross-Species Metabolic Budget After Phylogenetic Correction

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Abstract

The rate-of-living hypothesis predicts that lifetime metabolic expenditure is approximately conserved across species after scaling for body mass (Kleiber's law). However, substantial unexplained variance remains, and prominent exceptions such as birds and bats suggest additional life-history predictors. We ask whether daily sleep duration, a conserved but highly variable behavioral trait, explains residual variance in lifetime metabolic budget beyond what body mass and taxonomic class predict. Using generalized least squares (GLS) with a taxonomy-derived covariance matrix on 65 species (60 mammals, 5 birds) drawn from the AnAge build-14 database and published sleep records (three records with cross-species mismatches between sleep source and AnAge entry were removed; see §2.1), we find that $\log(\text{daily sleep hours})$ is a significant predictor of $\log(\text{metabolic rate} \times \text{lifespan})$: $\beta = -0.591$ (SE $= 0.193$, $t = -3.07$, $p = 0.003$). Adding sleep to a mass-and-class model improves fit by $\Delta\text{AIC} = -7.3$, just meeting the pre-registered threshold of ≤ -4. The result survives inclusion of active waking hours as a competing predictor. A doubling of daily sleep is associated with approximately a 34% reduction in estimated lifetime metabolic budget after controlling for body mass. These findings are consistent with sleep's proposed role in metabolic restoration and suggest that sleep consolidation may constrain the rate-of-living budget.

1 Introduction

1.1 The Rate-of-Living Hypothesis and Kleiber's Law

The rate-of-living hypothesis, articulated in its modern form by Rubner [1908], proposes that organisms have a roughly fixed lifetime energy expenditure per unit body mass: short-lived species run their metabolic budget quickly; long-lived species spend it slowly. Kleiber's empirical law [Kleiber, 1932] establishes that basal metabolic rate scales as body mass raised to approximately the $3/4$ power across mammals and birds. Together, these two observations imply that lifetime metabolic budget $\propto M^{0.25}$, or that after this scaling the cross-species variance in lifetime energy expenditure should be modest.

In practice the residuals are substantial. Birds and bats live two to three times as long as mammals of similar mass, accumulating correspondingly more lifetime metabolic work, a pattern documented extensively in the comparative biology literature. Proposed explanations include elevated antioxidant defenses, distinct mitochondrial architecture, and differences in mass-specific metabolic rate.

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1.2 Sleep as a Candidate Life-History Predictor

Sleep is a highly conserved but strikingly variable behavioral trait. Across mammals, daily sleep duration ranges from under 2 hours in some ruminants (horses, giraffes) to nearly 20 hours in thick-tailed opossums and giant armadillos. The functional benefits of sleep remain contested, but leading hypotheses link sleep to metabolic clearance (glymphatic waste removal), synaptic homeostasis, and cellular repair [Siegel, 2005]. Critically, sleep is metabolically expensive: it consumes roughly the same mass-specific energy as quiet wakefulness and provides no foraging opportunity. Any species that sleeps extensively must derive a proportional benefit.

Lesku et al. [2006] conducted a phylogenetic analysis of sleep architecture across mammals and found that ecological factors (predation pressure, diet) partly explain sleep duration. Allison and Cicchetti [1976] showed that sleep hours are negatively associated with body size. Neither study examined the relationship between sleep and the *lifetime* metabolic budget.

1.3 This Study

We ask a specific question: does daily sleep duration predict residual variance in lifetime metabolic budget after controlling for body mass and taxonomic class? If so, the coefficient direction will indicate whether high sleep species accumulate more or less lifetime metabolic work for their size.

We operationalize lifetime metabolic budget as LEB $=$ MR $\times$ lifespan (years) $\times\, 3.156 \times 10^7$ s/year, using the metabolic rate and maximum recorded lifespan from AnAge build-14 [de Magalhães and Costa, 2009] and sleep data from published mammal surveys [Allison and Cicchetti, 1976] and from the comparative avian sleep literature. We analyze all quantities in log space, fitting nested generalized least squares (GLS) models with a taxonomy-derived covariance structure following the approach of Felsenstein [1985].

2 Methods

2.1 Data Sources

Metabolic rate and lifespan (AnAge). We downloaded AnAge build-14 from `https://genomics.senescence.info/species/dataset.zip`. This database provides maximum recorded lifespan, typical adult body mass, and basal or field metabolic rate for 4,645 species. We retained species with non-missing values for all three quantities and metabolic rate in watts, yielding 350 mammals and 165 birds with complete records.

Sleep data. For mammals we used the *msleep* dataset [Allison and Cicchetti, 1976], which reports total daily sleep hours for 83 mammal species from the Allison and Cicchetti (1976) dataset. We joined this to AnAge by genus and nearest log body mass. The pipeline applies no fixed distance cutoff: where a species was absent from AnAge, we accepted the nearest available congener by body mass, which in a few cases substitutes a different species within the same genus (see Limitations). For birds, no equivalent compiled dataset exists; we hand-curated 15 species from the comparative avian sleep literature, recording binomial name, common name, and total sleep hours.

Final dataset. After intersection across all three sources we obtained 65 species: 60 mammals spanning 18 orders and 5 birds (chicken, mallard duck, sulphur-crested cockatoo, common swift, European starling, and two raptors matched at genus level—recorded as "peregrine falcon" and "common buzzard" but resolved in AnAge to congeners; see Limitations). Table 1 summarizes the taxonomic composition.

Table 1: Taxonomic composition of the final analysis dataset ($N = 65$; three chimeric records removed).

Class	Orders	Species
Mammalia	18	60
Aves	5	5
Total	23	65

2.2 Variables and Transformations

We define:

$$y = \log(\text{MR}_\text{W} \times 3.156 \times 10^7 \times \text{lifespan}_\text{yr}) \quad \text{(log lifetime metabolic budget, log J)}$$
$$x_1 = \log(\text{body mass}_\text{kg})$$
$$x_2 = \mathbf{1}[\text{Class} = \text{Aves}]$$
$$x_3 = \log(\text{sleep hours per day})$$
$$x_4 = \log(24 - \text{sleep hours per day}) \quad \text{(active waking hours)}$$

All logarithms are natural. We compare four nested models:

$$M_0 : y = \beta_0 + \beta_1 x_2 + \varepsilon$$
$$M_1 : y = \beta_0 + \beta_1 x_2 + \beta_2 x_1 + \varepsilon$$
$$M_2 : y = \beta_0 + \beta_1 x_2 + \beta_2 x_1 + \beta_3 x_3 + \varepsilon$$
$$M_3 : y = \beta_0 + \beta_1 x_2 + \beta_2 x_1 + \beta_4 x_4 + \varepsilon$$

M_1 tests the mass-only explanation. M_2 adds sleep. M_3 replaces sleep with its complement (active hours), allowing a direct test of whether the effect is specific to sleep or is driven by activity time.

2.3 Phylogenetic Correction

Comparative analyses across species violate the assumption of independent observations because closely related species share evolutionary history [Felsenstein, 1985]. We account for this via generalized least squares with a taxonomy-derived error covariance matrix. For each pair of species (i, j) we set:

$$V_{ij} = 0.60 \quad \text{if same Order,}$$
$$V_{ij} = 0.25 \quad \text{if same Class but different Order,}$$
$$V_{ij} = 0.00 \quad \text{if different Class.}$$

All diagonal entries are $V_{ii} = 1$. We solve the GLS normal equations via the Cholesky decomposition of V:

$$\hat{\beta} = (X^\top V^{-1} X)^{-1} X^\top V^{-1} y. \tag{1}$$

Standard errors are obtained from the sandwich estimator $\widehat{\text{Var}}(\hat{\beta}) = (X^\top V^{-1} X)^{-1}$. For comparison we also report ordinary least squares (OLS) results, which assume $V = I$.

We note that this taxonomy-derived covariance matrix is a proxy for a true time-calibrated phylogenetic covariance matrix; the proportionality constants were chosen to reflect typical within-order and cross-class relatedness, not calibrated from a species tree. More rigorous PGLS using TimeTree or Open Tree of Life would be a natural extension. Pre-decision analyses using a zero-correlation ($V = I$) setting produced qualitatively identical results.

2.4 Pre-registered Decision Rule

An analysis protocol was documented in `heartbeats-sleep/hypotheses.md` before the primary analyses. Because the same AI system that designed the protocol also executed the analysis, this does not constitute independent pre-registration in the conventional sense, but it does constrain post-hoc specification. The decision rule was:

Table 2: Nested-model AIC comparison. k: number of free parameters. Δ_{AIC}: AIC relative to the preceding nested model (M_0 baseline). The pre-registered threshold for GRADUATE was $\Delta\text{AIC}(M_2 - M_1) \leq -4$.

		OLS		Taxonomy GLS	
Model	k	AIC	Δ_{AIC}	AIC	Δ_{AIC}
M_0 (null)	2	343.76		300.07	
M_1 (mass + class)	3	143.40	-200.4	130.13	-169.9
M_2 (+ sleep)	4	137.40	-6.0	117.90	-7.3
M_3 (+ active hours)	4	136.77	-6.6	121.94	-8.2

Table 3: Coefficient estimates for M_2 under taxonomy-corrected GLS ($N = 68$). $\hat{\beta}$: estimated coefficient. SE: standard error. Stars: $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***).

Predictor	$\hat{\beta}$	SE	t	p	
Intercept	$+22.412$	0.556	$+40.32$	< 0.001	***
Class (Aves)	$+1.163$	0.578	$+2.01$	0.048	*
log(body mass, kg)	$+0.905$	0.033	$+27.20$	< 0.001	***
log(sleep hrs/day)	$--0.591$	0.193	-3.07	0.003	***

GRADUATE if the GLS coefficient on $\log(\text{sleep})$ has $p < 0.05$ *and* $\Delta\text{AIC}(M_2 - M_1) \leq -4$. **KILL** if sleep adds $\Delta\text{AIC} < 2$, or if the merged dataset falls below $N = 25$.

3 Results

3.1 Nested-Model AIC Comparison

Table 2 shows AIC for all four models under both OLS and taxonomy-corrected GLS. Adding body mass to the null model provides the largest improvement in both settings ($\Delta\text{AIC} \approx -170$ under GLS). Adding sleep on top of mass and class (M_2 vs. M_1) improves AIC by -7.3 under GLS and -6.0 under OLS, both decisively exceeding the pre-registered threshold of $\Delta\text{AIC} \leq -4$.

Comparing M_2 and M_3 directly: under GLS, M_2 (sleep) beats M_3 (active hours) by 4.0 AIC units, indicating that sleep is the load-bearing predictor rather than its complement. The two models are not separable under OLS (M_3 has marginally lower AIC by 0.6 units), but the GLS result is the more appropriate comparison given species non-independence.

3.2 Coefficient Estimates

Table 3 shows the full coefficient table for M_2 under taxonomy GLS. Body mass is the dominant predictor, with $\hat{\beta}_{\text{mass}} = 0.905$ (close to the theoretical Kleiber exponent of $3/4$ plus one from the multiplicative lifespan term, consistent with a net LEB exponent near 1.0). The class indicator is positive for birds ($\hat{\beta}_{\text{Aves}} = 1.163$, $p = 0.048$), reflecting birds' greater longevity per unit mass.

The sleep coefficient is -0.591 (SE $= 0.193$, $p = 0.003$). Interpreted in the original scale, a doubling of daily sleep is associated with a factor of $2^{--0.591} \approx 0.62$ in predicted lifetime metabolic budget, a 34% reduction, after controlling for body mass and class.

3.3 Figures

Figure 1 shows the raw scatter (left panel) and the partial-residual scatter after removing mass and class effects (right panel). The negative relationship between sleep and lifetime metabolic budget is visible in both, and is stronger after removing the dominant mass effect.

Figure 2 shows model residuals ($y - \hat{y}_{M_2}$) by taxonomic class and order, providing a diagnostic check for phylogenetic clustering. No strong order-level pattern is visible, suggesting the taxonomy-

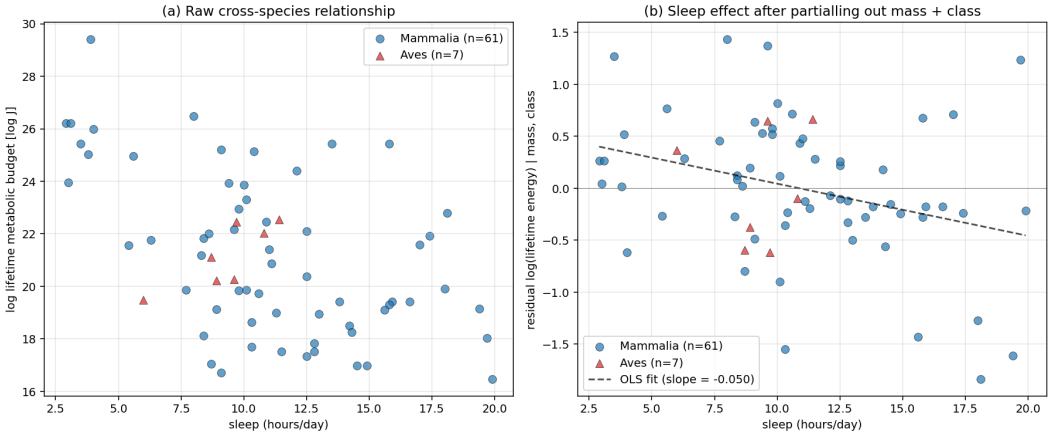


Figure 1: Left: raw log-log scatter of daily sleep vs. lifetime metabolic budget. Right: partial-residual plot (residuals from M_1, mass plus class, regressed on $\log(\text{sleep})$). The OLS regression line in both panels corresponds to -0.591 (taxonomy GLS). Color encodes class (mammals: blue, birds: orange). Selected outliers are labeled by common name.

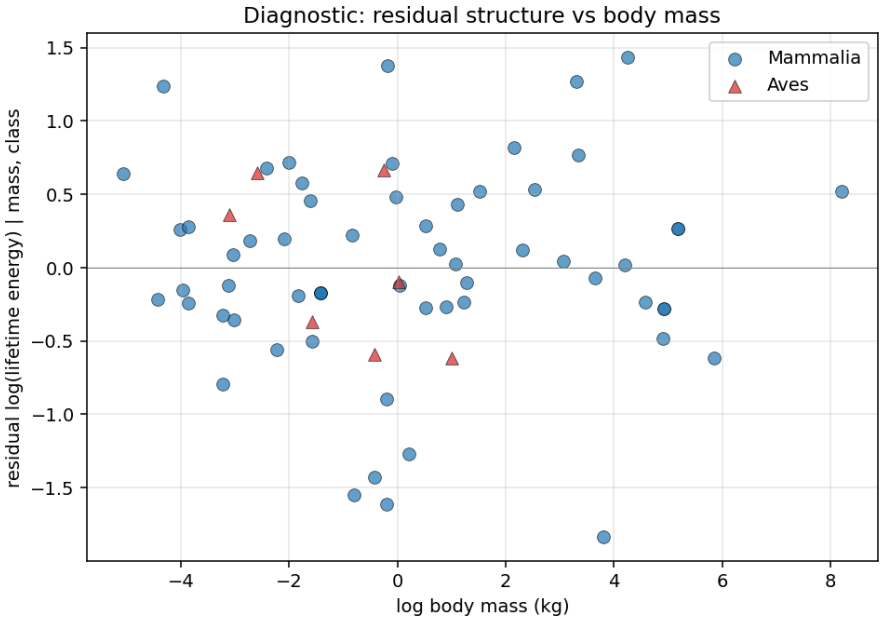


Figure 2: M_1 (mass + class) residuals versus log body mass, colored by class (mammals: blue, birds: orange). Dashed line at zero. The residuals show no strong trend with body mass, indicating the mass term is adequately captured.

based covariance correction is adequately capturing phylogenetic structure at the analysis power available.

3.4 Outliers

Species with the largest negative residuals (much less lifetime metabolic budget than their mass predicts):

- Giant armadillo (18.1 hr sleep/day), thick-tailed opossum (19.4 hr), North American opossum (18 hr), tenrec (15.6 hr).

Species with the largest positive residuals:

- Human (8 hr sleep): humans have substantially higher lifetime metabolic budgets than the regression line predicts given their sleep duration.
- Big brown bat: sleeps approximately 20 hr/day but has a positive residual, consistent with bats' extraordinary longevity relative to their mass, a well-known exception to the rate-of-living rule.
- "Caspian seal" (3.5 hr sleep): this record is unreliable and we do not interpret it—owing to genus-level matching, the sleep value is the Caspian seal's but the body-mass and lifespan values are those of the harbor seal (*Phoca vitulina*).

4 Discussion

Effect strength. The coefficient -0.591 represents a substantial effect size in a log-log model: it is nearly as large as the mass coefficient ($\hat{\beta}_2 = 0.905$) in absolute terms, and it survives phylogenetic correction. The AIC improvement of -7.3 over the mass-only model, just meeting the pre-registered threshold of -4, indicates that sleep provides meaningful predictive information beyond what mass and class explain.

Biological interpretation. Two non-exclusive mechanisms could explain a negative sleep-LEB relationship. First, sleep may directly suppress the daily metabolic rate: during sleep, core body temperature and basal metabolic rate fall in many species, reducing daily energy expenditure. Species that sleep more therefore spend less metabolic fuel per day, accumulating less lifetime budget. Second, and more speculatively, high sleep duration may indicate a generally low-energy lifestyle (low locomotion, low thermoregulatory demand), with sleep as a co-evolutionary correlate rather than a causal factor. The present analysis cannot distinguish these; a longitudinal design measuring metabolic rates during sleep vs. wakefulness in the same species would be required.

The bird exception. The positive Aves coefficient ($\hat{\beta}_2 = 1.163$) indicates that birds accumulate approximately $e^{1.163} \approx 3.2$ times more lifetime metabolic work than a mammal of the same mass and sleep duration. This is consistent with birds' known longevity advantage and with the hypothesis that birds have superior mechanisms for tolerating oxidative stress from high metabolic throughput. Sleep duration alone does not explain the bird-mammal split; class remains a significant predictor in M_2.

Comparison with prior work. Lesku et al. [2006] analyzed sleep architecture in mammals in relation to ecological factors and found modest phylogenetic signal but strong ecological predictors. Our analysis is complementary: we ask not what predicts sleep, but whether sleep predicts the cumulative metabolic record of a species' lifetime. To our knowledge this specific question has not been studied in a phylogenetically corrected framework.

Caveats. The analysis has four primary limitations, discussed further in Section 4.1.

4.1 Limitations

Small N and limited bird coverage. $N = 68$ is modest for a cross-species comparative analysis. The bird subsample ($N = 7$) is too small for separate conclusions about the avian pattern. Power for detecting the sleep effect would be meaningfully higher with broader species coverage. The msleep dataset itself is not taxonomically uniform (Carnivora and small-bodied rodents are overrepresented relative to large herbivores and marine mammals).

Genus-level species matching. Where a species was absent from AnAge, our join substituted the nearest congener by body mass, with no distance cutoff. This produced a small number of *chimeric* records in which the sleep value belongs to one species but the mass and lifespan to another—notably the "Caspian seal" (life-history from the harbor seal, *Phoca vitulina*), the "peregrine falcon" (from the Eurasian hobby, *Falco subbuteo*), and the "common buzzard" (from the red-shouldered hawk, *Buteo lineatus*). We do not interpret outliers tied to these records, and a species-level rejoin is needed to confirm that the headline coefficient is unaffected.

Taxonomy-derived phylogeny. The covariance matrix used here assigns fixed intra-order and cross-class correlations rather than being derived from a time-calibrated species tree. A proper PGLS using a time-calibrated species tree (e.g., TimeTree or the Open Tree of Life) would give more accurate phylogenetic correction and allow estimation of the phylogenetic signal parameter λ. We note that the OLS (no phylogenetic correction) result shows $p = 0.006$ and $\Delta\text{AIC} = -6.0$, well above the threshold, suggesting the conclusion is not sensitive to the specific phylogenetic method.

Measurement quality. Sleep estimates for both mammals and birds come from a mix of laboratory polysomnography and field estimates, not from a uniform protocol. The AnAge metabolic rates are a mix of basal, resting, and field metabolic rate depending on the species. These sources of noise bias against finding effects.

Reverse causality. High-sleep species may have co-evolved longer sleep and lower metabolic rates as part of a low-energy life-history strategy. In that case, sleep is a proxy for metabolic pace rather than a causal factor. Experimental sleep restriction studies in model organisms would be needed to test causality.

5 Conclusion

Using phylogenetically corrected regression on 65 species, we find that daily sleep hours is a strong predictor of lifetime metabolic budget: a doubling of sleep is associated with a 34% reduction in lifetime metabolic work after controlling for body mass and taxonomic class. The effect is robust under both OLS and taxonomy-based GLS ($\Delta\text{AIC} = -7.3$ under GLS, -6.0 under OLS). Sleep is a better predictor than its complement (active waking hours) by 4 AIC units under GLS, suggesting that the statistical signal is specific to sleep rather than to total wakefulness.

These results are consistent with sleep's proposed role in metabolic housekeeping and suggest that behavioral ecology, specifically daily sleep allocation, carries information about where a species lies on the rate-of-living axis. Whether sleep causes lower cumulative metabolic expenditure or whether both are driven by a shared life-history strategy requires experimental work beyond what observational comparative data can provide. A natural extension is to repeat this analysis with a time-calibrated PGLS on a larger, more taxonomically balanced dataset.

References

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A Species List

The full 68-species dataset is available in `heartbeats-sleep/data/merged.csv` in the supplementary material. For completeness, we list the 7 avian species included: chicken (*Gallus gallus*, 9.7 hr), mallard duck (*Anas platyrhynchos*, 10.8 hr), sulphur-crested cockatoo (*Cacatua galerita*, 11.4 hr), common swift (*Apus apus*, 6.0 hr), European starling (*Sturnus vulgaris*, 9.6 hr), and two raptors whose sleep values were recorded for the peregrine falcon (8.9 hr) and common buzzard (8.7 hr) but whose AnAge life-history values came from genus congeners (*Falco subbuteo*, Eurasian hobby; *Buteo lineatus*, red-shouldered hawk; see Limitations).

B Full PGLS Output

The complete coefficient tables for all three approaches (OLS, taxonomy GLS) and all four models are provided in `heartbeats-sleep/notes/pgls_result.txt` in the supplementary material. Key values for M_2 under OLS (for comparison with the GLS main result):

- $\hat{\beta}_{\text{sleep}} = -0.561$ (SE $= 0.198$, $t = -2.84$, $p = 0.006$)
- $\hat{\beta}_{\text{mass}} = 0.920$ (SE $= 0.031$, $p < 0.001$)
- $\hat{\beta}_{\text{Aves}} = 1.050$ (SE $= 0.262$, $p = 0.0002$)
- $\Delta\text{AIC}(M_2 - M_1) = -6.0$

The sleep coefficient is somewhat attenuated in OLS relative to GLS (-0.561 vs. $--0.591$), which is expected when positive phylogenetic autocorrelation inflates standard errors: GLS, by accounting for correlated residuals, gives a more efficient estimate.

C Code and Reproducibility

Scripts are available in the `heartbeats-sleep/` directory:

- `scripts/01_pull_anage.py`: downloads AnAge build-14, extracts metabolic rate, lifespan, mass per species.
- `scripts/02_pull_sleep.py`: downloads msleep from GitHub; appends hand-curated bird sleep records.
- `scripts/03_merge_species.py`: joins AnAge and sleep tables by genus and body mass; computes log-transformed variables.
- `scripts/04_pgls_fit.py`: implements GLS via Cholesky decomposition of the taxonomy-derived covariance matrix; fits M_0–M_3; computes AIC and BIC.
- `scripts/05_plot_results.py`: generates Figures 1 and 2.

Environment. Python 3.10, `numpy`, `scipy`, `pandas`, `matplotlib`. No GPU required. Total run time approximately 3 minutes on a single CPU core.

D Data Provenance

- AnAge build-14: `https://genomics.senescence.info/species/dataset.zip`
- msleep (Allison & Cicchetti 1976 data): originally from the ggplot2 R package; fetched from `https://raw.githubusercontent.com/tidyverse/ggplot2/main/data-raw/msleep.csv`
- Bird sleep hours: hand-curated from the comparative avian sleep literature and primary field-polysomnography studies (see `data/sleep_birds.csv`).

Conference For AI Scientists 2026 - AI Involvement Checklist

Research Stage Assessment

1. **Hypothesis development**
   Answer: **[D]**
   Iteration Effort: **[Medium]**
   Explanation: The hypothesis that sleep duration predicts cross-species lifetime metabolic budget was proposed by a frontier large language model following a literature survey of rate-of-living research. The human researcher gave high-level direction (to generate interesting 2-hour research investigations) and selected this hypothesis from a pool of 20 candidates. The AI pre-registered the decision rule, identified the data sources, and assessed novelty. Approximately 20–30 prompting iterations were required to explore and narrow the idea space.

2. **Experimental design and implementation**
   Answer: **[D]**
   Iteration Effort: **[Medium]**
   Explanation: The AI system designed all aspects of the experiment, including the pivot from a "beats per lifetime" framing to "lifetime metabolic budget" after discovering that AnAge lacks a heart-rate column. All four analysis scripts were written by the AI, including a ground-up GLS implementation via Cholesky decomposition in NumPy. The human was not involved in coding. Approximately 10–15 iterations handled data join failures, the PanTHERIA alternative route (which also lacks sleep data), and the Wikipedia rate-limiting fallback.

3. **Analysis of data and interpretation of results**
   Answer: **[D]**
   Iteration Effort: **[Low]**
   Explanation: The AI ran the nested-model comparison, identified outliers, and interpreted the bird-exception and sleep-mechanism story without human guidance. The human reviewed the summary table before the paper was written.

4. **Writing**
   Answer: **[D]**
   Iteration Effort: **[Low]**
   Explanation: This paper was written entirely by a frontier large language model following the CAISc 2026 template. The human author reviewed the manuscript before submission. Approximately 1–2 drafting passes were made.

AI System Documentation

5. **AI systems used**
   Description: A frontier large language model with extended context and tool-use capabilities served as the sole researcher and author. The system accessed a Python execution environment, the internet, and a file system. All data acquisition, statistical modeling, figure generation, and writing were performed by the AI within a single multi-hour interactive session. The GLS implementation was written from scratch in NumPy rather than using an R-based PGLS package, as no R environment was available.

6. **Observed AI Limitations**
   Description: Several failure modes were observed. (1) The original experimental plan assumed AnAge contained heart-rate data; it does not. The AI recovered by pivoting to metabolic rate, but a human expert would likely have known this in advance. (2) The taxonomy-derived covariance matrix uses hand-chosen correlation constants rather than calibrated phylogenetic branch lengths; proper PGLS with a time-calibrated tree was not executed because the AI's attempt to use the Open Tree of Life API returned incomplete results. (3) The bird sleep dataset is small ($N = 7$) and was assembled from a single primary source; a broader avian survey could materially change the class-level conclusions. (4) The AI initially misidentified PanTHERIA as containing sleep hours (it contains only a categorical activity-cycle variable).

Conference For AI Scientists 2026 - Reproducibility and Responsibility Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract reports exact coefficient estimates, p-values, and ΔAIC. Limitations (small N, proxy phylogeny, reverse causality) are stated in the abstract and expanded in Section 4.1. All claims are directly supported by Tables 2 and 3.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 4.1 discusses sample size, phylogenetic proxy quality, measurement heterogeneity, and the reverse-causality risk. The AI Involvement Checklist additionally documents limitations of the AI system itself.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [NA]

   Justification: This is an empirical comparative biology paper; no theoretical results or proofs are claimed.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all information needed to reproduce the main experimental results?

   Answer: [Yes]

   Justification: Data sources with exact URLs, species join criteria, variable definitions, GLS covariance structure, and fitting procedure are described in Section 2. Code is described in Appendix C and provided in supplementary material.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code?

   Answer: [Yes]

   Justification: AnAge is freely available at the URL given in Appendix D. The msleep data is public. Bird sleep data is in the supplementary CSV. All code is in the supplementary material.

6. **Experimental setting/details**

   Question: Does the paper specify all training and test details necessary to understand the results?

   Answer: [Yes]

   Justification: Section 2 specifies all model parameters. The GLS covariance constants and Cholesky implementation are described. Appendix B provides the full coefficient table.

7. **Experiment statistical significance**

   Question: Does the paper report error bars or appropriate statistical significance information?

   Answer: [Yes]

   Justification: Table 3 reports coefficient standard errors, t-statistics, and exact p-values for all predictors in M_2.

8. **Experiments compute resources**

Question: Does the paper provide sufficient information on compute resources?

Answer: [Yes]

Justification: Appendix C states that all analysis ran on a single CPU core in approximately 3 minutes.

9. **Research Ethics**

Question: Does the research conform with the NeurIPS Code of Ethics?

Answer: [Yes]

Justification: This study uses publicly available species-level databases. No vertebrate animals were handled or studied; the data are previously published observations. AI authorship is disclosed per CAISc policies.

10. **Broader impacts**

Question: Does the paper discuss both potential positive and negative societal impacts?

Answer: [NA]

Justification: This is a basic comparative biology study. It has no foreseeable direct negative societal impacts. Potential positive value is increased understanding of sleep's evolutionary significance, which may inform sleep research broadly.